CASCADIA ANALYTICS

Taiwan Semiconductor Manufacturing Company Limited

Equity Research Note | TSM | 21 OCTOBER 2026

AI Foundry Monopoly

Taiwan Semiconductor Manufacturing Co screens as a Hold, Valuation Concerns with a $243.42 blended target, 43% below the current price.

Analyst: Justin Wen

REFERENCE PRICE	BLENDED TARGET	BLENDED UPSIDE	
$425.08	$243.42	-43%	HOLD, VALUATION CONCERNS

Investment Thesis

AI Foundry Monopoly. TSMC manufactures the silicon inside Nvidia's AI accelerators. HPC is now 68% of revenue. Advanced nodes (7nm and below) are 79% of wafer revenue, with 2nm ramping from 3% to 8% in one quarter [8-K Q3 2026]. Demand Exceeds Supply. CEO C.C. Wei: AI demand still exceeds TSMC's supply capability. Q4 guided to $52.3-54.3B vs $48.8B consensus. Sixth consecutive record profit quarter [8-K Q3 2026]. Stretched vs DCF. At $425.08 [Mkt 16-Oct-26], the ADR trades 76% above our $241.71 DCF base. The

- AI Foundry Monopoly
- Demand Exceeds Supply
- Stretched vs DCF

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The latest print

Demand Exceeds Supply. CEO C.C. Wei: AI demand still exceeds TSMC's supply capability. Q4 guided to $52.3-54.3B vs $48.8B consensus. Sixth consecutive record profit quarter.

Stretched vs DCF. At $425.08, the ADR trades 76% above our $241.71 DCF base. The business is the best in semis; the price leaves no margin for geopolitical or cyclical error. All figures in NT$ unless noted.

NT$1,494.2B	NT$763.1B	66.4%	51.1%	68%	8%
REVENUE (Q3 2026)	NET INCOME (Q3 2026)	GROSS MARGIN (Q3 2026)	NET MARGIN (Q3 2026)	HPC REVENUE MIX	2NM WAFER MIX

Key metrics in detail

METRIC	LATEST	VS YEAR AGO	TREND
Revenue (Q3 2026)	NT$1,494.2B	+50.9%	Up
Net Income (Q3 2026)	NT$763.1B	+68.7%	Up
Gross Margin (Q3 2026)	66.4%	+6.9 pts	Down
Net Margin (Q3 2026)	51.1%	+5.8 pts	Up
HPC Revenue Mix	68%	+11 pts	Up
2nm Wafer Mix	8%	from 3% in Q2	Up
Revenue (FY25)	NT$3,809B	+31.6%	Up

The moat

AI demand exceeds supply; HPC now 68% of revenue (from 57%)

The growth engine

2nm ramp: 8% of wafer revenue (from 3% in Q2); advanced nodes 79%

Three scenarios make the valuation range explicit

Three scenarios, weighted 25% / 50% / 25% into the blended target.

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Taiwan Semiconductor Manufacturing Company Limited valuation framework

SCENARIO	WEIGHT	2026-30 CAGR	2030 REVENUE	2030 EBIT MARGIN	TARGET	UPSIDE
Bear	25%	9.0%	$248.44B	48.0%	**$198.89**	-53%
Base	50%	15.0%	$307.83B	52.0%	**$241.71**	-43%
Bull	25%	21.0%	$377.27B	56.0%	**$291.37**	-31%

Bear: downside case at $198.89 (-53%).

Base: central case at $241.71 (-43%).

Bull: upside case at $291.37 (-31%).

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The probability-blended target implies 43% downside

A 25% Bear / 50% Base / 25% Bull weighting implies -43% from the $425.08 reference price.

$243.42
BLENDED TARGET

The valuation uses a multi-year unlevered free cash flow DCF. The discount rate is the live CAPM rate: the 10-year Treasury yield plus beta times the market-implied equity risk premium. Terminal value is a Gordon perpetuity on terminal-year free cash flow.

Mechanics are explicit. The bridge runs revenue to EBIT to cash taxes, then adds back D&A and subtracts capex and working-capital investment to reach unlevered free cash flow. Targets are divided by diluted shares outstanding.

Revenue base (2025E)	**$176.00B**
Valuation horizon	**2026-2030**
Live WACC	**9.87%**
WACC path (Year 1 to final year)	**9.87% to 8.76%**
Terminal growth	**3.0%**
Net cash	**$55.30B**
2030 FCF (Base)	**$84.60B**
10-year Treasury yield	**5.1%**
Beta	**1.37**
Equity risk premium	**3.5%**

HOLD, VALUATION CONCERNS | $243.42 BLENDED TARGET | WEN

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The reverse DCF: what the market prices in

Holding the base-case FCF margin flat, the market-implied revenue CAGR is 29.2%. at/above Bull (21%) - euphoric

Reverse DCF sensitivity

Implied enterprise value and share price at each five-year revenue CAGR, with the base-case free cash flow margin held flat. The highlighted row sits nearest the market-implied CAGR.

CAGR	Implied EV	Implied price	CAGR	Implied EV	Implied price	CAGR	Implied EV	Implied price
+0%	$660.5B	$138.01	+20%	$1518.5B	$303.44	+40%	$3142.6B	$616.59
+1%	$690.5B	$143.80	+21%	$1578.4B	$314.99			
+2%	$721.6B	$149.80	+22%	$1640.3B	$326.92			
+3%	$754.0B	$156.03	+23%	$1704.1B	$339.23			
+4%	$787.5B	$162.49	+24%	$1770.0B	$351.93			
+5%	$822.2B	$169.20	+25%	$1837.9B	$365.03			
+6%	$858.3B	$176.15	+26%	$1908.0B	$378.54			
+7%	$895.6B	$183.35	+27%	$1980.3B	$392.47			
+8%	$934.4B	$190.81	+28%	$2054.8B	$406.84			
+9%	$974.5B	$198.55	**+29%**	**$2131.5B**	**$421.64**			
+10%	$1016.0B	$206.55	+30%	$2210.7B	$436.89			
+11%	$1059.0B	$214.84	+31%	$2292.2B	$452.61			
+12%	$1103.5B	$223.42	+32%	$2376.1B	$468.80			
+13%	$1149.5B	$232.30	+33%	$2462.6B	$485.47			
+14%	$1197.1B	$241.48	+34%	$2551.6B	$502.63			
+15%	$1246.4B	$250.97	+35%	$2643.2B	$520.30			
+16%	$1297.3B	$260.79	+36%	$2737.5B	$538.48			
+17%	$1349.9B	$270.94	+37%	$2834.6B	$557.19			
+18%	$1404.3B	$281.42	+38%	$2934.4B	$576.43			
+19%	$1460.5B	$292.25	+39%	$3037.0B	$596.23			

Balance sheet and share count reconciliation

Enterprise value converts to equity value after net debt, and targets divide by diluted shares.

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Taiwan Semiconductor Manufacturing Company Limited valuation framework

SCENARIO	ENTERPRISE VALUE	EXIT EV/ FCF	TERMINAL GROWTH	PLUS: NET CASH	EQUITY VALUE	TARGET
Bear	$976.32B	—	3.0%	$55.30B	$1,031.62B	**$198.89**
Base	$1,198.43B	—	3.0%	$55.30B	$1,253.73B	**$241.71**
Bull	$1,456.03B	—	3.0%	$55.30B	$1,511.33B	**$291.37**

Fair value across scenarios

Model fair value by year under each scenario, on the same paths the ticker page chart draws.

SCENARIO	2026	2026	2027	2028	2029	2030
Bear	$218.12	$218.12	$230.87	$243.67	$256.47	$269.10
Base	$265.07	$265.07	$282.32	$299.54	$316.44	$332.62
Bull	$319.52	$319.52	$342.65	$365.52	$387.42	$407.94

Methodology note. The discount rate is the live U.S. 10-year Treasury yield plus beta times equity risk premium. Terminal growth reflects long-run nominal GDP. The $243.42 headline target applies 25% / 50% / 25% Bear / Base / Bull weights.

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Taiwan Semiconductor Manufacturing Company Limited valuation framework

The franchise deep dive

AI demand exceeds supply; HPC now 68% of revenue (from 57%)

2nm ramp: 8% of wafer revenue (from 3% in Q2); advanced nodes 79%

Q4 guide $52.3-54.3B vs $48.8B consensus; full-year growth ~44% in USD

Revenue and margin trend

METRIC	FY21	FY22	FY23	FY24	FY25
Revenue (NT$B)	NT$1,587	NT$2,264	NT$2,162	NT$2,894	NT$3,809
Gross Margin	51.6%	59.6%	54.4%	56.1%	59.8%
Net Margin	37.9%	44.9%	39.0%	40.0%	44.6%

Valuation cross-check

NTM P/E 17.6x

DCF Base $241.71 vs $425.08 spot

PEG 0.8

PEG is attractive, but DCF implies 43% downside. The multiple is fair; the absolute price is not.

Foundry peers trade at lower multiples but lack leading-edge share

Premium to peers is justified by technology leadership.

Trading near all-time highs

Above historical multiple range on AI enthusiasm.

What to monitor

Jan 2027. Q4 2026 Earnings: $52.3-54.3B guide (vs $48.8B Street). 2nm ramp is the watch item.

2027. 2nm volume ramp and Arizona fab expansion progress.

2026-2028. CapEx guided significantly above prior 3-year period to meet AI demand.

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Financial summary: income statement

YEAR	REVENUE	GROSS PROFIT	OPERATING INCOME	EBITDA
2021	$1587.4B	$819.5B	$650.0B	—
2022	$2263.9B	$1348.4B	$1121.2B	$1593.1B
2023	$2161.7B	$1175.1B	$921.4B	$1523.5B
2024	$2894.3B	$1624.4B	$1322.0B	$2079.1B
2025	$3809.1B	$2281.3B	$1936.1B	$2742.1B

Balance sheet

YEAR	TOTAL ASSETS	TOTAL LIABILITIES	TOTAL EQUITY	CASH
2021	$3725.3B	$1573.6B	$2149.3B	$1065.0B
2022	$4964.5B	$2046.6B	$2917.8B	$1342.8B
2023	$5532.2B	$2078.3B	$3453.9B	$1465.4B
2024	$6691.8B	$2412.5B	$4279.3B	$2127.6B
2025	$7932.8B	$2536.6B	$5396.2B	$2767.9B

Cash flow

YEAR	OPERATING CF	CAPEX	FINANCING CF	FREE CF
2021	$1112.2B	$-839.2B	$136.6B	$273.0B
2022	$1610.6B	$-1089.6B	$-200.2B	$521.0B
2023	$1242.0B	$-955.4B	$-204.9B	$286.6B
2024	$1826.2B	$-965.0B	$-346.3B	$861.2B
2025	$2275.0B	$-1282.6B	$-440.3B	$992.4B

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Agency risk score 52 of 100: balanced, typical corporate alignment

The score follows the website's Executive Alignment & Agency Risk model: insider ownership up to 40 points, open-market Form 4 flow over the trailing six months up to 30, and pay-for-performance up to 30. Insiders hold 0.01% of shares outstanding. Form 4 filings over the trailing six months show $24.4K of open-market purchases against $0 of sales, a net flow of $24.4K. One-year total shareholder return is +47.1% against CEO total pay of $63.23M (Dr. C. C. Wei Ph.D.), which the model scores as PERFORMANCE EARNED.

52
AGENCY RISK SCORE, OF 100

The three components

COMPONENT	SCORE	WHAT THE SITE SHOWS
Insider Ownership (Skin in the Game)	2 / 40	0.01% of shares held by insiders
Form 4 Flow (Conviction Flow)	24 / 30	Buys $24.4K (4) vs sales $0 (0), net $24.4K, trailing six months
Pay-for-Performance	26 / 30	PERFORMANCE EARNED: Stock Outperformance Matches Comp Structure. 1Y TSR +47.1%; Dr. C. C. Wei Ph.D., Chairman & CEO, total pay $63.23M
Agency Risk Score	**52 / 100**	Regime: BALANCED / TYPICAL CORPORATE

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Recent Form 4 filings

DATE	FILER	RELATION	TYPE	SHARES	VALUE
2026-07-28	Tien Bor-Zen	Officer	BUY	15	$5.8K
2026-07-20	Tien Bor-Zen	Officer	BUY	10	$4.0K
2026-05-20	Tien Bor-Zen	Officer	BUY	17	$6.7K
2026-04-28	Tien Bor-Zen	Officer	BUY	20	$7.8K

SEC proxy audit

N/A. The weekly DEF 14A proxy pipeline holds no CD&A audit record for this ticker (foreign private issuers file no DEF 14A), so no audit findings are shown.

Methodology. This module reuses the website's Executive Alignment & Agency Risk model unchanged. Scores of 75 and above band as founder-mode, high conviction; 45 to 74 as balanced, typical corporate; below 45 as high agency risk. Inputs are the same feeds behind the site model (Yahoo Finance insider holdings, Form 4 transactions and executive compensation) plus the SEC DEF 14A CD&A audit from the weekly proxy pipeline. Computed 21 October 2026 for this note.

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The bear case

Geopolitical Risk

Concentration of leading-edge fabs in Taiwan; any cross-strait escalation is existential for the equity.

Node Transition

2nm ramp diluted Q3 gross margin by 1.3 pts; yield issues at new nodes are the recurring execution risk.

Customer Concentration

Customer concentration (Apple, Nvidia) means any share shift or demand pause is material.

The strengths

Operating Quality

The world's leading foundry with no close second at the leading edge. 66% gross margins, 51% net margins, NT$2.77T cash, 0.19x debt/equity. Pricing power is structural.

Recent Trend

Sixth straight record quarter. Profit +68.7% YoY, revenue +50.9%. Q4 guide raised well above consensus. Demand exceeds supply per CEO.

Forward Growth

2nm ramping fast, Q4 guide implies +58% YoY, 2026-2028 capex stepped up materially. AI accelerator demand is the multi-year driver.

Valuation

At $425.08 vs $241.71 DCF base, the ADR is 76% above fair value, the premium behind our Hold, Valuation Concerns rating. PEG 0.8 is attractive on growth, but absolute price embeds flawless execution.

Cascadia scorecard

Composite score 76: Attractive. Confidence High, as of 17-Oct-2026.

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PILLAR	RATIONALE	SCORE
Operating Quality	The world's leading foundry with no close second at the leading edge. 66% gross margins, 51% net margins, NT$2.77T cash, 0.19x debt/equity. Pricing power is structural.	95
Recent Trend	Sixth straight record quarter. Profit +68.7% YoY, revenue +50.9%. Q4 guide raised well above consensus. Demand exceeds supply per CEO.	95
Forward Growth	2nm ramping fast, Q4 guide implies +58% YoY, 2026-2028 capex stepped up materially. AI accelerator demand is the multi-year driver.	85
Valuation	At $425.08 vs $241.71 DCF base, the ADR is 76% above fair value, the premium behind our Hold, Valuation Concerns rating. PEG 0.8 is attractive on growth, but absolute price embeds flawless execution.	40
Catalysts & Risk	Q4 earnings and 2nm yield progress are positives. Taiwan geopolitical risk is the permanent overhang. Rates-driven semis selloffs hit the stock regardless of fundamentals.	70

Investment view

A 2nm yield problem or Taiwan escalation would cut all pillars. Sustained 40%+ growth into 2027 with margin recovery could support current levels.

Investment view: Hold, Valuation Concerns. The $243.42 blended target implies -43%, balancing the $198.89 / $241.71 / $291.37 Bear / Base / Bull outcomes.

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